



Mozilla launches Firefox Send

Mozilla officially released Firefox Send, a file sharing service that uses end-to-end encryption to keep shared files private. <https://dgit.in/apr19-93>



Samsung's 'perfect full-screen'

Samsung is working on making the entire front of its phones a screen, with selfie cameras under the display. <https://dgit.in/apr19-94>

over the years, while the chips themselves have been shrinking in size. The major providers for smartphone flash memory chips are the same as the DRAM chips – Samsung, Micron and Hynix. Two other companies, Toshiba and Western Digital Corporation (WDC) also have a significant share in the market. At times, the same devices can use NAND chips from multiple manufacturers. The Samsung Galaxy S10 for example, uses both Toshiba and Samsung NAND flash memory chips.

CAMERA

The camera sensors are provided primarily by two manufacturers, Samsung under the ISOcell brand name, and Sony under the Exmor

this, which interested readers can read up about in the February 2019 issue of Digit. HTC has bucked the trend by using UltraPixel sensors, which have individual pixels more than twice the size of offerings by Samsung and Sony, using the unconventional approach of improving image quality by just letting in more light. Qualcomm's imaging head, Mr Judd Heape has indicated that smartphone camera sensors breaching the 100MP barrier will be announced later in the year, and make their way into smartphones by 2020.

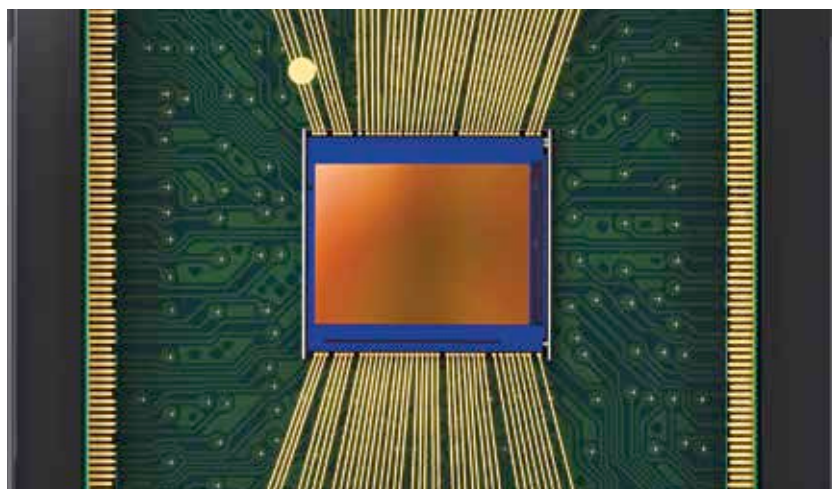
SCREENS

The screens are made up of a display covered by a protective glass coating. While there are many display technologies, the top two in common use

LG. Companies may use displays from either manufacturer. Pixel devices, for example, have used LG displays for one variant, and Samsung displays for another variant of their phones.

Then there is the glass protection, which are mostly from two major suppliers for all smartphones. Corning provides its Gorilla Glass, while Asahi sells Dragontrail glass. There are no major differences between the two, but western companies prefer Corning. Corning is an America-based company, while Asahi is based in Japan. Both Gorilla Glass and Dragontrail Glass are alkali-aluminosilicate glass, and provide roughly the same amount of resistance to fall and scratch damage. Some very rare devices – such as the HTC U Ultra use a sapphire crystal protection for the outermost layer. This sapphire crystal is not glass at all, is much more resistant to scratches than either Gorilla Glass or Dragontrail, but is much more prone to shattering.

The Samsung Galaxy S10 comes with its own plastic screen protector out of the box, as not all third party protectors work well with the ultrasound in screen fingerprint scanner. Some users have found out that the response from the scanner is better if the screen is actually removed. Samsung is offering a superior glass screen protector with its phones.



The 20 MP Samsung ISOCELL Slim 3T2 sensor

brand name. Apart from the topline offerings, most of the sensor offerings by both the companies are practically identical, allowing smartphone manufacturers to use either one of the two for sourcing in a production run of a particular device. Along with the other smartphone components, the general trend has been to reduce the size of the image sensors. At the same time, the increasing megapixel count means that each imaging element on the sensor becomes smaller, and this leads to a deterioration in imaging quality as the amount of light reaching each element is reduced. There are some smart ways that companies have worked around

are active-matrix organic light-emitting diode (AMOLED), and In-Plane Switching Liquid Crystal Display (IPS LCD). The higher end devices have AMOLED screens, while the mid range and entry level devices have IPS LCD screens. AMOLED displays are brighter, support a wider range of viewing angles, have better contrast, and are a newer technology. IPS LCD screens are cheaper, but AMOLED screens are expected to spread to more lower end devices. Samsung dominates the AMOLED market, with a staggering 84% market share, which actually came down from 89% in 2017. The other big display manufacturer is

BATTERIES

There are a handful of manufacturers that provide most of the batteries by volume. These companies are Samsung, Toshiba, Panasonic and LG Chem. However, as lithium-ion batteries are not that hard to manufacture, there are many small companies involved that are not among the top suppliers. Even the large technology companies may source batteries from smaller companies that are not that well known, and are not the top suppliers worldwide. Smartphone batteries are poised to get increasingly expensive over the years, because of competition from the Automobile sector. Electric vehicles use lithium-ion batteries too, and there are some rare